Exploring the Potential of Generative AI: Use Cases in Software Startups

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Abstract. *Background and Related Work*: Software startups face unique challenges in product development, including limited resources, the need for rapid innovation, and the constant pressure to adapt to market changes. Generative Artificial Intelligence (GenAI) has recently gained significant attention, offering capabilities to assist creative processes, generate content, and enhance decision-making through data analysis. However, how GenAI can be integrated into agile product development processes in software startups remains an open question. *Objective*: This study aims to identify potential use cases for GenAI in software startups and explore how GenAI can support innovation, overcome development challenges, and integrate with agile practices to improve product quality and development speed. *Method*: We identified a list of GenAI use cases from existing systematic literature reviews and mapped them to engineering process areas in software startups. Following that, we conducted workshops with experts to validate our results. *Results*: The results provide a descriptive overview of GenAI's potential applications in software startup environments. Given the current state of the art, we identified areas that could benefit faster from integrating GenAI. *Conclusions*: The study delineates the prospective impact of GenAI on agile product development in software startups, showcasing areas of immediate applicability.

Keywords: software startups · product development · potential applications · generative ai

1 Introduction

Given their flexibility and agility, software startups have enormous potential to disrupt markets and innovate rapidly [5]. However, they face unique challenges in product development, including limited resources, the need for rapid innovation, and the constant pressure to adapt to market changes [7]. Agile methodologies are widely adopted in software startups to address these challenges, as

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they enable iterative development, continuous feedback, and quick adaptation to evolving requirements [9].

The ongoing revolution in generative artificial intelligence (GenAI) technologies may offer promising features for software startups to enhance their product development processes, accelerate innovation, and improve product quality [6]. As discussed by Tripathi et al. [9], choosing technologies to speed up product development is crucial for software startups.

GenAI refers to technologies that can create content of various forms based on a large-sized trained dataset [1]. These technologies have shown significant potential across various industries, including software development and product innovation [6]. Although the potential of GenAI in software startups is promising, existing research has yet to explore how GenAI can be integrated into engineering processes at startups. Consequently, We developed the following research question to guide us in conducting this study.

What are the potential use cases of GenAI to support software startups in their software engineering practices and challenges?

This study is our first effort to explore and use GenAI in software startups. Specifically, we aimed to identify potential use cases for GenAI capabilities in the product development in software startups. To achieve this goal, we first identified the type of GenAI technologies and their capabilities from the academic literature. Then, we mapped these capabilities with the engineering model for software startups. As a result, we identified use cases for potential uses of GenAI for software startups.

This paper's contributions include a list of potential use cases of GenAI mapped to the engineering process and avenues to continue research with GenAI in software startups. The remainder of this paper is organized as follows. The next section provides an overview of software startups, GenAI technologies, and their potential applications in software startups. Section 3 presents the research methodology used in this study. Section 4 presents the results of our research and the potential use cases for software startups. Section 5 discusses the implications of our findings. Finally, Sect. 6 concludes the paper and summarizes our key findings and avenues for future research.

2 Related Work

2.1 Software Startups

Software startups are newly established companies that develop innovative software products or services under challenging circumstances [10]. The primary goal of startup companies is to craft a scalable and sustainable business [8]. Studies report that software startups often partially adopt unorthodox software development practices during product development due to the influence of diverse factors, including entrepreneurial vision, market needs, and capital constraints [7,9]. This emphasizes the essential role of adaptable methodologies like Lean Startup and Agile, often present in software startups [7].

Klotin et al. [4] identify common goals, practices, and challenges across life-cycle and engineering process areas in the software startup context. The life-cycle consists of four stages: inception, stabilization, growth, and maturity. The inception stage focuses on forming a capable team and developing a minimum viable product (MVP). Stabilization involves refining the product based on customer feedback and preparing for scaling. Growth emphasizes achieving market share, while maturity focuses on optimizing operations and maintaining market position. The engineering areas include team, requirement engineering, value focus, quality goals and testing, architecture and design, and project management. The study outlines several key challenges and practices in software startups. Challenges include forming and managing a skilled team, particularly in the early stages, dealing with dynamic requirements, avoiding feature creep, and maintaining quality assurance while managing technical debt. To address these, startups adopt practices such as using external experts for specialized tasks, relying on iterative and agile development methodologies to adapt to rapid changes, and establishing continuous feedback loops with customers to validate requirements.

2.2 GenAI and Its Potential

GenAI represents a subfield of artificial intelligence focused on synthetic content generation [1]. It employs advanced generative models, large language models (LLMs), trained on vast text, graphics, audio, or video datasets [1]. These models analyze patterns and relationships within the datasets to synthesize novel responses to user prompts [6]. GenAI technologies hold significant promise across various industries, potentially fostering new ideas, automating tasks, enhancing creative exploration, streamlining processes in software engineering, and beyond [2]. These vast potentials of GenAI attract new ventures[1] and projects[2].

A comprehensive study groups more than 350 GenAI applications into 15 use categories, encompassing text, images, video, 3D, code and software, speech, AI understanding, business, gaming, music, biotech, brain, and others, showcasing the technology's versatility [3]. Zhang et al. categorize GenAI by the type of content it generates, such as text, images, video, 3D, and speech, while also exploring the technology's industrial applications [11].

3 Methodology

Our methodology leverages existing systematic literature reviews to explore the application of GenAI in software startups:

1. GenAI Use Cases: We referred to (two) existing systematic literature reviews: one by Zhang et al. [11], which covers various GenAI content generation tasks, and another by Zheng et al. [12], focusing on the application of LLMs in software engineering tasks. These reviews provide a representative snapshot of

[1] https://www.ycombinator.com/companies/industry/ai.
[2] https://github.com/filipecalegario/awesome-generative-ai.

advancements and applications in GenAI. We selected these two systematic literature reviews because they cover a broad range of GenAI use cases, offering a clear view of current trends and research focus areas. Given the field's rapid pace, we aim to capture its direction and key topics of interest.
2. Practices and Challenges in Software Startups: We referred to the progression model of software engineering in startups by Klotins et al. [4] to select specific goals, challenges, and practices across startup stages mapped into six engineering process areas.
3. Cross-Referencing and Synthesis: Independently, each author cross-referenced GenAI use cases (step 1) with the startup progression elements (step 2). We then converged to synthesize a collective analysis, addressing contradictions through discussion.
4. Expert Workshop for Validation: We presented our results to two senior researchers twice to validate our synthesis, followed by additional meetings to discuss and refine the analysis. The senior researchers are experts in software engineering in the software startup context.

4 Results

Below we present our results as the ***Use cases*** (), derived by mapping the elements of the progression model () [4] for software startups to the use cases identified in the literature reviews [11,12]. These results directly answer our research question by showing how the identified use cases align with the six elements of the deployed progression model [4], highlighting the practical applications of GrnAI in addressing software engineering challenges in startups.

Team area refers to forming and managing effective teams [4]. **Goals** include establishing a team with sufficient skills and expertise, recognizing the team as a catalyst for product development, and emphasizing characteristics like cohesion, coordination, leadership, and continuous learning for project success. **Challenges** involve team formation, management, expertise, leadership, coordination, and engineering skills shortages. **Practices** in this area include establishing a feedback loop with customers, documenting feature ideas, and determining the "good-enough" level of quality, all aimed at advancing through the startup life-cycle stages and addressing the difficulties in practicing software engineering.

Use cases: In the team area, GenAI can support cross-team communication, such as automatic time zone tracking with multi-lingual translation, generating and managing communication artifacts (e.g., emails), visualizing data and ideas, creating presentations, transcribing, and summarizing discussions. Additionally, trained chatbots can support various engineering activities such as generating source code, developing relevant test cases, and fixing bugs. Moreover, GenAI can also help recruit new team members and communicate with external stakeholders in a personalized way.

Requirement engineering area focuses on eliciting, analyzing, validating, documenting, and scoping software requirements [4]. This area is crucial for identifying and validating relevant product ideas, which are essential startup activities. The primary **goal** of this area is to identify market opportunities, devise feasible solutions, and prioritize customer feedback to drive product development and innovation. **Challenges** include transitioning from inventing requirements to using customer input as the product matures, managing feature creep, and maintaining domain knowledge through documentation. **Practices** involve collecting input from potential customers early on to identify relevant requirements, analyzing the usefulness of features to customers to prevent feature creep, and documenting requirement ideas to facilitate knowledge sharing and avoid misunderstandings within the team.

Use cases: GenAI can play a role in the requirements engineering area, e.g., acting as a user with specific characteristics (data-driven persona) to generate requirements, refine ideas from various use cases, and identify key features. Furthermore, it can help the team capture, document, analyze, and synthesize immersive feedback given in any language from customers around the globe. A trained GenAI can guide the team to develop meaningful MVPs based on comprehensive feedback, relevant test cases, and documentation.

Value focus area consists of several **goals** that are related to maintaining a focus on value [4]. These include enhancing the external and customer-perceived value through better functionality and user experience, realizing the internal market potential to capture and expand market share, generating significant financial value through effective revenue management, and establishing internal differentiation value to set the product apart from competitors. There is no specific **challenge** in this area.

Use cases: GenAI can offer tools to enhance user experience and streamline development. By simulating user interactions, GenAI can perform preliminary quality checks, identifying potential issues and areas for improvement before a product reaches its final stages. This capability not only improves the user experience but also accelerates value delivery. Additionally, GenAI-driven chatbots can be instrumental in guiding decision-making processes related to technology selection. These chatbots analyze vast amounts of data and user inputs to tailor recommendations on the most suitable technologies and frameworks.

Quality goals and testing area consists of goals, challenges, and practices related to quality and testing [4]. The **goals** include ensuring the functionality of the product to meet specific needs, maintaining maintainability for easy updates and repairs, achieving a quick time-to-market to stay competitive, enhancing portability to function across different platforms, and determining a "good-enough" quality level that balances performance with cost-effectiveness. The **challenge** involves manual regression testing, which requires significant effort. **Practices** involve eliciting and validating quality requirements to align product specifications with user needs, conducting informal, manual, exploratory

testing to uncover unforeseen issues, and establishing a robust QA process with clearly defined roles for testers.

Use cases: GenAI can offer valuable insights into optimal interaction pathways and support usability studies, which is essential for refining user interfaces. It also plays a role in generating test cases and test scenarios, enhancing test automation processes. Additionally, it can provide recommendations for defining the criteria necessary to achieve MVP quality, ensuring that the product meets essential requirements while remaining lean. Moreover, GenAI assists in collecting and analyzing performance metrics, suggesting improvements for software products, and offering code corrections to increase coding quality. Furthermore, chatbots powered by GenAI can be useful in guiding manual testing efforts, and their capabilities extend to automating these tests to reduce the manual work in test execution.

Architecture and design refers to the technical decisions and design choices that shape the product's structure and functionality [4]. Startups prioritize stable technologies over cutting-edge alternatives to minimize risks and accelerate product development. They often adopt open-source and well-established frameworks to avoid reinventing the wheel. **Challenges** are related to managing technical debt, especially as startups scale, with earlier architecture decisions leading to complications. For UI design, startups utilize mockups and design frameworks, continuously improving based on customer interaction data, aiming for usability and usefulness. **Practices** focus on using best practices and avoiding complex, untested solutions, with some mature startups innovating in-house to support product evolution.

Use cases: In addressing architecture and design challenges, task-oriented GenAI can help the startup select suitable frameworks and generate architecture diagrams to enhance the development process and stakeholder comprehension. Descriptive image generation enables the creation of user interface mockups directly from textual descriptions, streamlining the design phase. For technical debt mitigation, GenAI for code completion and automated test generation accelerates development and enhances quality and maintainability.

Project Management within software startups involves strategically planning and controlling resources to navigate from inception to maturity [4]. It is crucial for optimizing resource usage and achieving set milestones. **Challenges** include the lack of clear, measurable goals in the early stages and evolving needs for more sophisticated metrics for progress assessment. Resource constraints and the necessity for flexible adjustment to market demands further complicate project management. **Practices** evolve from an informal approach, often based on intuition, to more structured methods using planning and resource management tools as startups mature.

Use cases: In project management, GenAI technologies may play a role in enhancing strategic planning and resource control. Task-oriented chatbots support collecting and analyzing metrics, facilitating quick planning adjustments,

and aiding in project management activities such as meetings and budgeting. Moreover, using GenAI can schedule resources, prioritize tasks, and predict future needs based on data analysis. These potential uses contribute to startups' transition from intuition-based methods to structured, data-informed strategies, streamlining the process from inception to maturity and addressing challenges and the need for flexible market adjustment.

5 Discussion

Our results show that GenAI can support software startups in various use cases during product development across life-cycle and engineering areas [4]. Most of GenAI's use cases are carried out through interactive conversational chatbots, which allow interaction using various forms of data [6]. Popular use cases include multi-lingual translation, code and test generation, access to expert and knowledge systems, and image generation. These capabilities enable more efficient development, better alignment with customer needs, and continuous process improvement.

However, not all GenAI use cases are equally relevant to software startups. For example, no suitable use cases for features like generating 3D talking characters, complex 3D models, and music generation. Moreover, there is no clear connection between GenAI features and the following product development goals: balancing customer value with time-to-market considerations, internal market potential value, financial value, and revenue generation. We believe these complex goals demand human judgment and domain expertise beyond the current capabilities of GenAI.

We identified various GenAI use cases broadly supporting general work-related tasks. These include capabilities such as summarizing information, generating content, and facilitating communication. We identified tailored use cases related to software tasks that directly contribute to product development, such as generating test cases and providing architectural recommendations. These applications demonstrate how GenAI can enhance efficiency and decision-making within software development processes. However, the overall impact and value of GenAI tools for software startups remain to be thoroughly evaluated. Future studies are necessary to determine how these technologies can best be integrated into the startup ecosystem and to what extent they can replace or augment existing processes.

6 Conclusion

Software startups have unique characteristics, goals, challenges, and engineering practices. On the other hand, the recent advancement of GenAI brings opportunities to help the software industry and beyond. This study aimed to identify potential use cases for GenAI technologies in software startups and explore how these technologies can support innovation and overcome development challenges to improve the product development process [9].

Our mapping shows use cases where GenAI can help software startup during product development spread in six engineering areas. However, several product development goals are not directly addressed using GenAI.

Future research directions will focus on validating our findings through empirical studies. We also aim to develop practical guidelines for startups on effectively integrating GenAI into their product development processes. This will involve simulating scenarios, experimenting with ideas, validating assumptions, etc.

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